Compositional vs. Conceptual Novelty in Text-to-Image Models

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Abstract

This paper presents an empirical study of diffusion-based text-to-image models and investigates how their creative capacity can be evaluated through the novelty of generated images. Inspired by parallels to human dreaming, we investigate whether the apparent originality of these systems reflects genuine conceptual creativity or merely the recombination of patterns learned from training data. Specifically, this work aims to assess whether diffusion models produce truly novel visual representations or operate within the constraints of their learned distributions.

To evaluate image's novelty across different complexity levels, three types of complex prompts were implemented and used with **Stable-XL Diffusion** model, direct (in-distribution), compositional (multi-concept recombination), and abstract (conceptually non-concrete or semantically stressed). For each generated image, we computed semantic alignment using **CLIP** similarity, distributional deviation via embedding-based novelty metrics, and output variability through diversity measures. Statistical analysis, including one-way ANOVA and confidence intervals, was applied to evaluate differences between abstraction levels and to demonstrate results are not due to chance.

The results indicate that Stable-XL Diffusion model showed variations in diversity and novelty metrics over different prompt levels. However, the model operates primarily through constrained recombination rather than true conceptual extrapolation, supporting the hypothesis that artificial generative systems -like human dreams- reconstruct rather than invent representations beyond their experiential manifold.

This study has several limitations. Novelty is measured using embedding-based metrics, which provide only an indication of distributional deviation rather than definitive conceptual originality. Furthermore, the analysis focuses on a single diffusion-based model, and the results may not generalize across different architectures or training datasets. Since the exact training data distribution is unknown, conclusions about true extrapolation remain inferential.

Keywords—diffusion models, Stable-xl Diffusion model, novelty, CLIP similarity, human dreams.

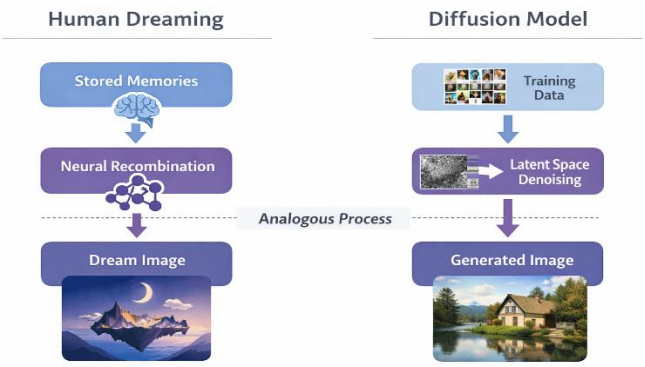
I. Introduction

Generative text-to-image models are described as creative systems capable of producing novel visual content from arbitrary textual descriptions. Synthetic images generated by these models now circulate widely across digital platforms, often appearing indistinguishable from human-created or real-world imagery. Despite this rapid advancement and widespread adoption, a fundamental question remains unresolved: is the novelty observed in generated images conceptual in nature, or is it primarily compositional emerging from the recombination of previously learned elements?

In this work, we explore the conceptual and computational relationship between human dreaming and diffusion-based image generation models. Dreams are often understood as a recombination of stored memories into unfamiliar configurations [10]. Although dream experiences feel novel, they are constructed from prior representations. Inspired by this analogy, our study investigates whether the generative behavior of diffusion models can be analyzed through a framework inspired by dream-like visual generation, focusing particularly on the balance between semantic alignment and visual novelty.

Several recent studies have explored different aspects of generative AI evaluation, including image quality assessment, semantic alignment between prompts and generated images, and the measurement of novelty in generative outputs. In addition, research on multimodal representation learning has introduced powerful tools such as contrastive language-image models that enable quantitative evaluation of generated content. However, most of them primarily focus on technical evaluation metrics and do not explore conceptual parallels between human cognitive processes and artificial generative mechanisms.

Our work introduces a novel perspective that integrates the concept of human dreams into the evaluation of generative diffusion models. By analyzing the generated images using multimodal similarity metrics and novelty estimation techniques, we aim to investigate how diffusion models balance creativity and semantic consistency. The central research question guiding this study is: **To what extent do diffusion-based generative models produce genuinely novel visual concepts, and how similar is this process to the compositional and associative mechanisms observed in human dreaming?**



II. Related Work

Liu et al. (2022) propose Composable Diffusion Models, factorizing an image into components by combining multiple diffusion modules (object-level, relation-level, etc.) as energy-based generators. They show this structured approach can produce scenes more complex than the training set, effectively composing novel arrangements of entities. Our work instead uses a standard diffusion model without modular decomposition; we analyze the natural compositionality in its outputs rather than creating a special architecture. Nevertheless, Liu et al. demonstrate that diffusion models can, in principle, represent highly compositional images when properly guided, highlighting the potential for diffusion outputs to contain compositional novelty. [1]

Wu et al. (2024) propose ConceptMix, an evaluation benchmark for T2I compositionality. ConceptMix automatically generates text prompts containing multiple visual concepts (using GPT-4 for prompt generation) and uses a vision-language model to verify how many concepts appear in each generated image. They find that as prompt complexity (number of concepts) increases, model performance drops sharply, especially for open models. This highlights current limits of diffusion models in fully capturing compositional prompts. In our work, we similarly test models on multi-concept prompts, but we also examine the nature of novel elements (conceptual vs compositional) in the images, comparing them to how humans might dream up combinations. [2]

Ramaswamy et al. (2024) focus on quantifying creativity in generative vision models. They formalize three criteria **(**prompt satisfaction, output cohesion, and diversity**)** to iteratively measure how models introduce new elements. Their metrics align with intuitive notions: high creativity means new elements that are cohesive with prompt. This is directly relevant to our novelty analysis. We adopt similar concepts by measuring how diffusion outputs introduce new visual content relative to prompt constraints. Ramaswamy et al.'s framework confirms that diversity and cohesion can be computed numerically, validating our use of CLIP similarity and other diversity statistics. [3]

Deperrois et al. (2023) explicitly link dreams with modern machine learning concepts. They propose that dreaming may implement "adversarial dreaming", a feedback/feedforward learning game akin to GAN training, and "contrastive dreaming" that enforces invariance through mapping similar dream experiences. These frameworks suggest the brain uses internal imagery (dreams) as training signals. While our work is empirical, it is thematically connected: we examine whether diffusion model generations, which combine learned data patterns, show signs of such constructive or contrastive mixing. In particular, adversarial and contrastive processes in dreaming motivate our analysis of whether generated images blend concepts in novel ways. [4]

Our work differs by empirically comparing diffusion-generated images to human dream analogues. We analyze whether the novelty in diffusion outputs is **compositional** (built from prompt components) or **conceptual** (emergent surprising elements), paralleling how dreams recombine experiences. Unlike prior work that optimizes for creativity, we measure inherent novelty in unconstrained outputs. Overall, our study contributes a novel analysis framework linking machine generation to cognitive dream processes.

Three main hypotheses emerged from our review of previous related works.

H1 (generative modeling as generalization of experience), the generative model is assumed to produce only what can be expressed through concepts it has encountered in the training data. This is similar to human dreams, which only "dream" of what our senses have previously experienced. **Liu et al. (2022) [1]** supports this idea; they showed that large-scale diffusion models (such as DALLE-2) generate very realistic images but often fail to understand or integrate new elements that did not appear together during training. In other words, the model remains within its own sphere of experience and only reconstructs familiar concepts.

H2 (the limits of novelty: combination, not invention), we assume that a model can combine familiar elements to create new scenes, but without inventing entirely new elements outside of the training experience. **Liu et al. (2022)[1]** also confirms this premise; they proposed generating the scene by synthesizing multiple diffusion models, each with a specific component (such as an object or property), and showed that this method generates scenes far more complex than any single example in the training. In other words, the model creates new scenes by combining, for example, "cat + chair = cat on a chair," but it does not invent elements from scratch.

H3 (direct recall versus generation), we acknowledge that models may sometimes retrieve almost identical familiar images from the training data. We haven't found a specific study in the literature that directly supports this, but it's similar to the phenomenon of fully dreaming about a previously seen scene. Some community observations suggest that generative models may sometimes produce exact replicas (not documented here), thus limiting their "imagination."

III. Model selection & Prompts levels

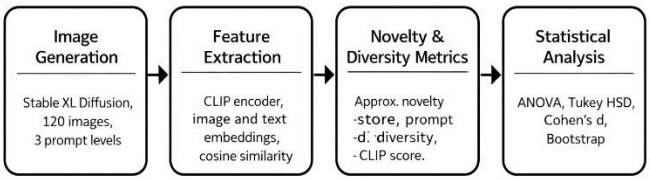
To investigate generative novelty, we selected **Stable XL Diffusion[5]**, a latent diffusion-based text-to-image model. Stable Diffusion was chosen due to its widespread adoption, open-source accessibility, and strong performance in generating high-quality images from textual descriptions. As a diffusion model trained on large-scale image-text pairs, it learns a structured latent representation of the visual distribution, making it particularly suitable for studying whether generated outputs remain within the learned manifold or extend beyond it. Its architectural transparency and reproducibility also make it appropriate for systematic empirical evaluation of novelty and abstraction effects.

To examine how novelty behaves under varying semantic demands, prompts were organized into three levels of abstraction. **Level 1 (Direct Prompts)** consisted of concrete and commonly encountered descriptions, representing in-distribution generation. **Level 2 (Compositional Prompts)** combined multiple known concepts to test the model's ability to recombine learned elements in novel configurations. **Level 3 (Abstract Prompts)** introduced conceptually non-concrete or semantically stressed descriptions designed to push the limits of the model's representational capacity. This structured progression enables a controlled comparison between compositional novelty and potential conceptual novelty under increasing abstraction.

IV. Methodology

A. Experimental Design:

To investigate the nature of generative novelty in diffusion-based text-to-image systems, we designed an experimental framework that combines controlled prompt generation with quantitative embedding-based evaluation. The objective was to examine how varying levels of semantic abstraction influence the behavior of a generative model in terms of semantic alignment, diversity of outputs, and distributional novelty.[8]



The experimental pipeline consists of four principal stages:

- Image generation using a diffusion-based model.
- Feature extraction using multimodal embeddings.
- Computation of novelty and diversity metrics.
- Statistical evaluation of differences across prompt abstraction levels.

All experiments were conducted using the text-to-image diffusion model **Stable Diffusion**, while semantic alignment and embedding representations were obtained using **CLIP**.

B. Image Generation:

Images were generated from textual prompts using Stable Diffusion. The model produces images through an iterative de-noising process within a latent representation space learned from large-scale image-text datasets.

For each prompts level, we defined **5** different prompts, then asked the model to generate **8** images for each prompt, so that for each level we had **40** images, with total **120** images across three levels. By structuring prompts across three levels, the experiment enables a controlled analysis of whether novelty increases through recombination or through genuine conceptual extrapolation, and to ensure that the model does not duplicate images from the same prompt but generates new image each time.

To ensure experimental consistency, the following conditions were maintained across all generations:

- Identical inference parameters.
- Identical sampling procedure.
- Fixed generation pipeline.
- Different random seeds for each image sample.

C. Semantic Alignment Measurement:

Semantic alignment between generated images and their corresponding prompts was evaluated using **CLIP** embeddings. The **CLIP (**Contrastive Language Image Pre-Training) model is a multimodal deep learning model developed by **OpenAI** that learns to connect images and text within a shared embedding space. It is trained on large-scale image-text pairs and can measure semantic similarity between a textual description and an image representation.[6]

For each generated image:

- The image was encoded using the CLIP image encoder.
- The textual prompt was encoded using the CLIP text encoder.
- Cosine similarity between the two embeddings was computed.

The resulting similarity score measures the degree to which the generated image corresponds semantically to its textual description (prompt). Higher similarity values indicate stronger alignment between visual output and textual input.[9]

D. Diversity Measurement:

To measure variability among images generated from the same prompt, we computed **prompt diversity** using pairwise cosine distances between CLIP image embeddings.[9]

For each prompt:

- Embeddings of all generated images were extracted.
- Pairwise cosine distances were computed.
- The average distance was used as the diversity score.

This metric reflects how much the model varies its outputs when conditioned on the same prompt.[11]

E. Approximate Novelty Metric:

To estimate generative novelty, we computed an embedding-based metric referred to as **approximate novelty**. This metric measures how far generated images deviate from a reference distribution in embedding space. As a reference distribution, we used embeddings derived from the **MS COCO**[7], a widely used large-scale dataset containing diverse real-world images.

The procedure was implemented as follows:

- **2000** Images from the MS COCO dataset were encoded using the CLIP image encoder to construct a reference embedding distribution representing typical visual concepts.
- Each generated image was encoded into the same CLIP embedding space.
- The distance between the generated image embedding and the reference distribution centroid was computed.
- This distance was used as an **approximate novelty score**, representing the degree of deviation from common visual representations.

Although this metric cannot directly measure conceptual originality, it provides a quantitative proxy for how far generated outputs deviate from typical real-world visual distributions.

F. Statistical Analysis:

To determine whether abstraction level significantly influences the measured metrics, several statistical analyses were performed:

- **One-way ANOVA** to test overall differences between prompt levels
- **Tukey HSD post-hoc tests** for pairwise comparisons
- **95% confidence intervals** for each metric
- **Cohen's d effect sizes** to quantify magnitude of differences
- **Bootstrap estimation** to assess stability of mean values
- **Correlation analysis** to examine relationships between metrics

All analyses were conducted using standard scientific Python libraries including NumPy, SciPy, and statistical packages.

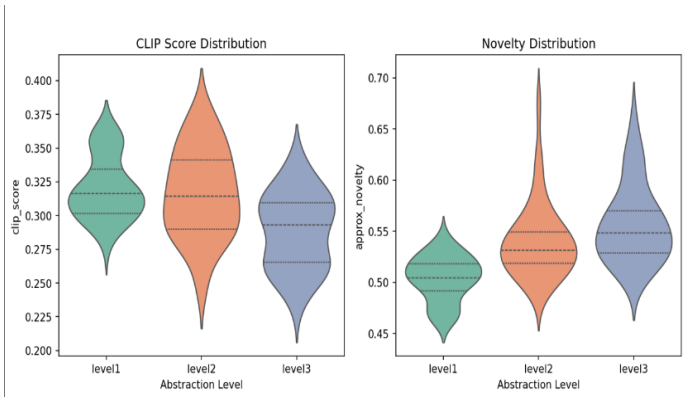
V. Results & Discussion

A. Approximate Novelty Analysis:

The approximate novelty scores ranged between **0.46** and **0.49** for the sampled outputs, indicating that the generated images are neither direct reproductions of known visual patterns nor entirely unrelated to the learned distribution. Instead, the results suggest that the model produces outputs that remain partially anchored to the learned feature space while still exhibiting moderate novelty. This behavior aligns with the assumption that diffusion models generate images through recombination of previously learned representations rather than generating entirely unseen visual structures.

B. Prompt Complexity Trend:

A clear trend emerged across the three prompt levels. As the conceptual complexity of prompts increased from Level 1 to Level 3, **the CLIP score gradually decreased while the approximate novelty increased**. This indicates that more abstract prompts push the model further away from its learned visual distribution. However, this shift also leads to weaker alignment between the generated image and the textual description.



[Figure 1]: *Distribution of metrics across abstraction levels*

C. Confidence Interval Analysis:

Metric	Level	Mean	95% CI
CLIP Score	Level 1	0.3205	[0.31, 0.33]
	Level 2	0.3167	[0.30, 0.33]
	Level 3	0.2886	[0.28, 0.30]
Approx. Novelty	Level 1	0.5041	[0.50, 0.51]
	Level 2	0.5403	[0.53, 0.55]
	Level 3	0.5570	[0.54, 0.56]

[Table 1]: *95% Confidence Intervals*

The confidence intervals for the evaluated metrics were relatively narrow, suggesting that the measured values are statistically stable and unlikely to result from sampling noise. In particular, the approximate novelty intervals show consistent separation between the prompt levels, reinforcing the reliability of the observed trends.

D. ANOVA Results:

Metric	F-value	p-value
CLIP Score	15.25	< 0.001
Prompt Diversity	48.52	< 0.001
Approx. Novelty	26.99	< 0.001

[Table 2]: *One-Way ANOVA*

The one-way ANOVA tests indicate statistically significant differences across the three prompt levels for all evaluated metrics (CLIP score, prompt diversity, and approximate novelty). This confirms that variations in prompt abstraction significantly affect the behavior of the generative model, validating the experimental design used in this study.

E. Effect Size Analysis:

Metric	Comparison	Cohen's d	Interpretation
CLIP Score	Level1 vs Level2	0.132	Small
	Level1 vs Level3	1.245	Large
	Level2 vs Level3	0.923	Large
Novelty	Level1 vs Level2	-1.209	Large
	Level1 vs Level3	-1.704	Large
Diversity	Level1 vs Level2	-2.558	Large6

[Table 3]: *Effect Sizes*

Effect size measurements (Cohen's d) further support the significance of the observed differences. Large effect sizes were observed particularly between Level 1 and Level 3 in both **clip_score** and **approx_novelty**. This indicates that the differences are not only statistically significant but also practically meaningful in terms of model behavior.

F. Post-Hoc Pairwise Comparisons:

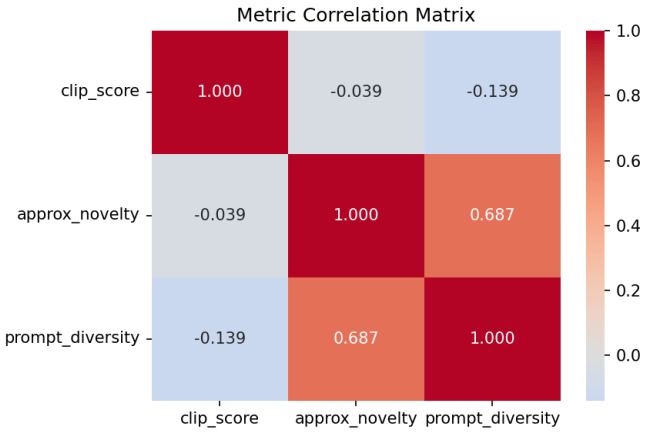
Metric	Comparison	Mean Difference	p-value	Significant
CLIP Score	Level1 vs Level2	-0.0037	0.8247	No
	Level1 vs Level3	-0.0318	<0.001	Yes
	Level2 vs Level3	-0.0281	<0.001	Yes
Diversity	Level1 vs Level2	0.0597	<0.001	Yes
	Level1 vs Level3	0.0931	<0.001	Yes
	Level2 vs Level3	0.0334	0.002	Yes
Novelty	Level1 vs Level2	0.0362	<0.001	Yes
	Level1 vs Level3	0.0529	<0.001	Yes
	Level2 vs Level3	0.0167	0.065	No

[Table 4]: *Tukey HSD Results (Selected)*

Post-hoc analysis reveals that the most substantial differences occur between **Level 3** and the other levels. While **Level 1** and **Level 2** show relatively small differences in some metrics, **Level 3** consistently produces significantly different outcomes. This suggests that high levels of conceptual abstraction trigger a distinct generative behavior in the model.

G. Correlation Analysis:

Table 6*: Correlation Matrix*



[Figure2]: *Metric correlation matrix*

The correlation analysis shows a strong positive relationship between prompt diversity and approximate novelty. This suggests that increasing semantic diversity within prompts directly contributes to higher novelty in generated images. Conversely, the weak correlation between CLIP score and novelty indicates that creativity and semantic alignment operate as partially independent dimensions in generative models.

H. Bootstrap Estimation:

Metric	Level	Mean	Bootstrap SE
CLIP Score	Level 1	0.3206	0.0036
	Level 2	0.3170	0.0048
	Level 3	0.2885	0.0044
Novelty	Level 1	0.5041	0.0035
	Level 2	0.5403	0.0057
	Level 3	0.5568	0.0060
Diversity	Level 1	0.0422	0.0014
	Level 2	0.1021	0.0051
	Level 3	0.1350	0.0099

[Table 5]*: Bootstrap Estimates*

Bootstrap resampling was used to evaluate the statistical stability and reliability of the estimated metrics. Since the experiment relies on a relatively small number of generated images, bootstrap allows us to repeatedly sample from the existing data and estimate how much the calculated means may vary. The resulting bootstrap standard errors are very small across all metrics, indicating that the estimated values for CLIP score, novelty, and prompt diversity are stable and not highly sensitive to the specific sample used.

I. Interpretation:

Overall, the results suggest that generative diffusion models exhibit a behavior similar to cognitive processes associated with human dreaming. Rather than generating entirely new visual concepts, the model appears to synthesize novel outputs by recombining elements from previously learned representations. As prompts become more abstract and conceptually demanding, the generated images deviate further from typical training patterns, increasing novelty while reducing semantic alignment. This behavior supports the hypothesis that generative models produce "creative" outputs through structured recombination within their learned representation space, rather than through the creation of fundamentally new visual knowledge.

VI. Conclusion & Future Work

This study investigated whether novelty in images generated by diffusion models represents true **conceptual** creativity or simply **compositional** recombination of learned patterns. By evaluating generated images under progressively abstract prompt conditions, the results revealed a clear pattern: as prompt abstraction increased, the measured novelty and diversity of the generated images also increased, while semantic alignment with the prompt slightly decreased. These findings suggest that diffusion models are capable of producing outputs that appear increasingly novel when faced with abstract or unfamiliar descriptions. However, this novelty emerges primarily through recombination and transformation of learned visual concepts rather than the creation of entirely new representations.

Despite providing useful insights, this study has several **limitations**. First, the experiment was conducted using a relatively small set of generated images and prompts, which limits the generalizability of the results. Second, the novelty measurements relied on proxy metrics such as CLIP similarity and embedding distance rather than direct access to the model's original training distribution. Finally, the evaluation focused on a single diffusion model, which means the conclusions may not fully represent the behavior of other generative architectures.

Future research can expand this framework by evaluating larger datasets of generated images and incorporating additional generative models for comparison. Access to known training datasets or partial training distributions could enable more precise measurements of novelty relative to the original data. Furthermore, future studies may explore more advanced evaluation metrics, including human perceptual studies and multimodal embedding analyses, to better understand how artificial systems generate seemingly new visual concepts and how this process relates to cognitive theories of imagination and dreaming.

References

[1] Compositional Visual Generation with Composable Diffusion Models

[2] ConceptMix: A Compositional Image Generation Benchmark with Controllable Difficulty

[3] Quantitative Measures of Task-Oriented Creativity in Popular Generative Vision Models

[4] Learning beyond sensations: How dreams organize neuronal representations

[5] Stable diffusion XL model

[6] CLIP (Contrastive Language Image Pre-Training)

[7] Microsoft COCO: Common Objects in Context

[8] High-Resolution Image Synthesis with Latent Diffusion Models

[9] Learning Transferable Visual Models From Natural Language Supervision (CLIP)

[10] The Brain as a Dream State Generator: An Activation-Synthesis Hypothesis of the Dream Process

[11] Evolve to Inspire: Novelty Search for Diverse Image Generation

Code, evaluation framework and generated images on GitHub repository

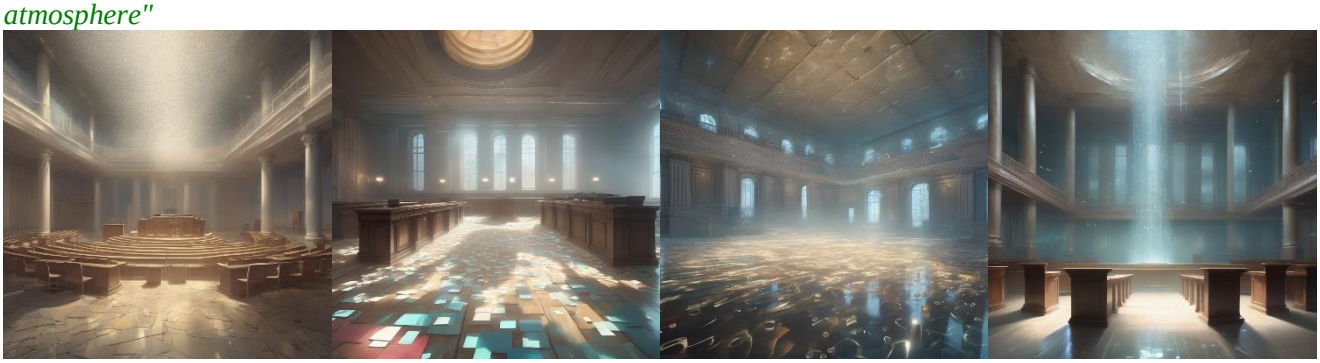
Appendix:

A.1 Direct Prompts (Level 1)

Example prompt:
"A cinematic photograph of an old fisherman standing on a wooden dock at sunrise, wearing a weathered blue raincoat and rubber boots, holding a fishing net full of silver fish, with seagulls flying overhead, gentle waves reflecting golden sunlight, distant mountains covered in light morning mist, highly detailed textures, natural lighting, shallow depth of field"

A.2 Compositional Prompts (Level 2)

Example prompt:
"A transparent glass elephant filled with glowing bioluminescent jellyfish, walking through a flooded Renaissance cathedral, sunlight refracting through stained glass windows, reflections on water, floating candles, cinematic volumetric lighting, ultra-detailed textures"

A.3 Abstract Prompts (Level 3)

Example prompt:
"A courtroom where abstract concepts like justice, fear, and probability appear as physical entities made of light and glass, debating in a space where the floor is a flowing ocean of numbers, ultra-realistic textures, cinematic atmosphere"